ch2 Heat Capacity

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2025 Q1

1. An equal amount of energy is supplied to each of the substances *P*, *Q*, *R* and *S* of mass and specific heat capacity listed below. Which substance would have the largest rise in temperature ?

	substance	**mass / kg**	**specific heat capacity / J kg^{-1} $^{\circ}$C^{-1}**
A.	*P*	2	4200
B.	*Q*	3	2100
C.	*R*	4	900
D.	*S*	5	840

2017 Q1

1. 30 g of milk at 10°C is added to 120 g of coffee at 80°C. Assuming there is no heat loss to the surroundings, what is the final temperature of the mixture ?

Given: specific heat capacity of milk = 3800 J kg^{-1} $°C^{-1}$

specific heat capacity of coffee = 4200 J kg^{-1} $°C^{-1}$

A. 64.8°C
B. 65.2°C
C. 66.0°C
D. 67.1°C

2022 Q1

1. A well-insulated vessel with negligible heat capacity contains warm water at 45 °C. The temperature of warm water drops by 5 °C after mixing with 50 g water at temperature 0 °C added to the vessel. Find the original mass of warm water in the vessel.

   A. 400 g
   B. 450 g
   C. 500 g
   D. 550 g

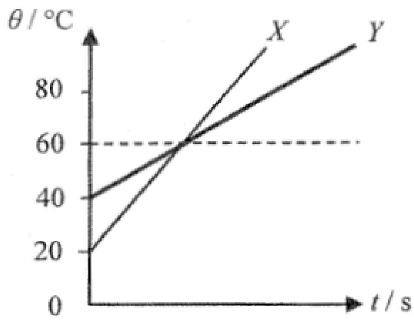
2019 Q1

1. A block measuring 80 °C is put into water at a temperature of 40 °C. The final temperature of the mixture is 70 °C. Which of the following deductions must be correct ? Assume that there is no heat loss to the surroundings.

   A. The energy gained by the water is greater than the energy lost by the block.
   B. The mass of the water is greater than the mass of the block.
   C. The specific heat capacity of water is smaller than that of the block's material.
   D. The heat capacity of the water is smaller than that of the block.

2015 Q2

2.



Two objects X and Y are made of the same material. They are heated separately by heaters of the same power. The graph shows the variation of temperature θ of X and Y with time t. What is the ratio of mass of X to that of Y ?

A. 3 : 1
B. 2 : 1
C. 1 : 2
D. 2 : 3

2020 Q2

2. An electric kettle which contains 1 kg of water at room temperature takes 168 s to heat up the water to boiling point. The kettle's rated value is '220 V, 2000 W'. Assume that all the electrical energy consumed by the kettle is transferred to the water. Which of the following statements is/are correct ? Given: specific heat capacity of water = 4200 J kg^{-1} $^{\circ}C^{-1}$

(1) The initial temperature of the water is 20 °C.
(2) The resistance of the kettle's heating element is about 24 Ω.
(3) If the electric kettle is operated with 110 V, the time taken to heat up the water to boiling point will be doubled.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1), (2) and (3)

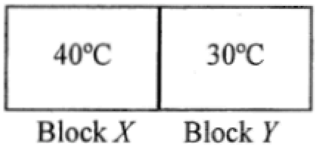
3. Milk at 5 °C is added to a cup of tea at 25 °C. Which statements below are correct ? Neglect the heat capacity of the cup and assume that there is no heat exchange with the surroundings.

(1) The average kinetic energy of the water molecules in the tea decreases.
(2) The average potential energy of the water molecules in the tea remains unchanged.
(3) The energy lost by the tea equals the energy gained by the milk.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2012 Q1

1. Two metal blocks X and Y of the same mass and of initial temperatures 40°C and 30°C respectively are in good thermal contact as shown. The specific heat capacity of X is greater than that of Y. Which statement is correct when a steady state is reached ? Assume no heat loss to the surroundings.



A. The temperature of block X is higher than that of block Y.
B. Their temperature becomes the same and is lower than 35°C.
C. Their temperature becomes the same and is higher than 35°C.
D. Their temperature becomes the same and is equal to 35°C.

2016 Q2

2. 0.3 kg of water at temperature 50 °C is mixed with 0.2 kg of ice at temperature 0 °C in an insulated container of negligible heat capacity. What is the final temperature of the mixture ?

Given: specific heat capacity of water = 4200 J kg^{-1} °C^{-1}

specific latent heat of fusion of ice = 3.34×10^5 J kg^{-1}

A. −1.8 °C

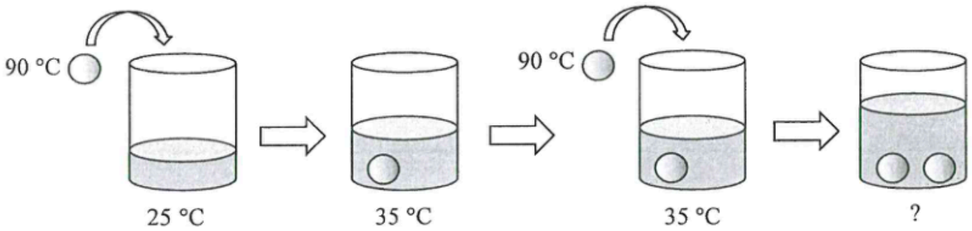
B. 0 °C

C. 1.8 °C

D. 3.0 °C

2026 Q3

3.



A metal sphere at 90 °C is put into a cup of water. The temperature of the water increases from 25 °C to 35 °C. If a second identical metal sphere, also at 90 °C, is added to the water, what is the final temperature of the water ? Assume there is no heat loss to the cup or the surroundings throughout the process.

A. 41.5 °C
B. 42.3 °C
C. 43.5 °C
D. 45.0 °C

pp Q2

2. In the figure below, a training pool B is located next to the main pool A. The training pool B has a smaller area and is shallower. If the pools are under the sunlight at the same time, which of the following statements about the rise in the water temperature of the two pools is correct ? Assume that the initial water temperatures of the pools are the same.

A. The water temperature of training pool B rises faster because it is shallower.
B. The water temperature of training pool B rises faster because it has a smaller surface area.
C. The water temperature of main pool A rises faster because it is deeper.
D. The water temperature of main pool A rises faster because it has a larger surface area.

pp Q3

3. Peter adds 50 g of milk at 20°C to 350 g of tea at 80°C, what is the final temperature of the mixture ?

Given : Specific heat capacity of milk = 3800 J kg^{-1} °C^{-1}
Specific heat capacity of tea = 4200 J kg^{-1} °C^{-1}

A. 50.0°C
B. 72.5°C
C. 73.1°C
D. 77.4°C